

Presenter Reference Notes – September 2026

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Slide 1: NDNF: A Data-Centric Service Framework for

Secure, Collaborative, and Dynamic Networked Applications Department of Computer Science University of Memphis Advisor: Dr. Lan Wang PhD Proposal Defense September 2026

Slide 2: Start with a Task: Several Views, One Result

An operator selects a ground area. Nearby UAVs offer suitable views. A compute Provider analyzes the images. The operator receives original images with annotations. UAV A view 1 UAV B view 2 Compute Provider Operator Who can participate now, and which Data may each selected Provider use?

Slide 3: The Task Creates Three Requirements

1. Discover current willingness: reachability, sensors, load, and deadlines affect participation. 2. Authorize before execution: exposing a service does not grant everyone permission to use it. 3. Enforce the chosen collaboration: an authentic image is useful only if it satisfies an assigned dependency. The contribution must explain the mechanism and its cost, not just demonstrate that two UAVs can exchange images.

Slide 4: What NDN Provides—and What It Does Not

NDN building block Remaining application responsibility Named Data and signatures [1] Decide which signer may perform which task; validate content under trust rules Interest retrieval and caching [1,26] Enforce task identity, dependencies, and deadlines; caching is not persistent storage Dataset synchronization / SVS [2,3] Discover new names, then retrieve Data; define service-message state transitions NAC [4] / enhanced NAC-ABE [19] Relate decryption permission to execution authorization and key lifecycle NAC: Name-based Access Control; ABE: attribute-based encryption; SVS: State Vector Sync. NDNF builds on these mechanisms; it does not invent them.

Slide 5: Three Research Questions

RQ1: How can a requester discover and authorize currently willing Providers without making the discovery descriptor public, and at what cost? RQ2: How can Providers selected at runtime follow task roles and named dependencies, whether Data comes from producers or caches? RQ3: When do these mechanisms improve completion, latency, or execution efficiency, and when do their costs outweigh the benefits? UAV and distributed inference test the same questions. They are not separate contributions merely because both are implemented.

Slide 6: Related Work Already Provides Important Pieces

Prior work What the comparison must acknowledge NFN [6], NFaaS [7] Named computation and function execution/placement RICE [9], NSC [10] Invocation and security mechanisms; NSC validates signed calls

using application trust rules DNMP [12] Role- and command-constrained signing and verification CaS-CON [15] Service-composition research; requires a feature-level comparison before claiming a new composition mechanism gRPC [17,27] Complete configurations can include discovery/resolver integration, not only fixed addresses Candidate increment: connect runtime assignment to enforceable object-admission rules. This is not a claim to be the first NDN service framework.

Slide 7: Contributions and Evidence Are Different Things

Intended contribution Current evidence boundary Discovery / authorization design and tradeoffs Invocation implementation and bounded experiments exist; full alternative lifecycle-cost comparison remains open ACK-derived plan and local dependency admission Design and implementation work; complete direct/cache and failure acceptance gates remain open Cross-case empirical findings Existing mobility/work-efficiency and small-model results support limited configurations, not all new collaboration claims Proposal status, September 2026: implemented, locally tested, measured, and fully qualified are not interchangeable. [28]

Slide 8: Participants and Trust Bootstrap

User: requests and coordinates the invocation. Provider: offers and executes a service. A node may play both roles. Controller: administers identities and service permissions approved by the organization. Participants use provisioned trust anchors and validation rules for certificates and Controller material. Unknown runtime candidates are not unregistered authorization identities. The User discovers current participation; the Controller does not choose this task's Providers.

Slide 9: One Invocation, Four Named Data Types

Data Logical purpose Request Describe required service and participation constraints; include protected discovery information and Requester Challenge ACK Report positive or negative participation; positive ACK supplies selection information and Provider Challenge Selection Authorize selected execution; provide input/references and protected request-scoped keys Response Return status, result Data, or authorized result references Sync helps discover names; Interests retrieve the Data. A positive ACK does not itself start execution.

Slide 10: Descriptor Before Selection; Input After Selection

Discovery descriptor Needed to decide whether to participate: area, deadline, sensor/model and resource requirements. Application input Needed to execute the selected task: images, detailed processing parameters, or protected input references. The boundary follows necessity. A location needed to assess coverage belongs in the protected descriptor; an image needed only for processing can wait.

Slide 11: RQ1: Why Protect Discovery Content?

The User knows the service, but not yet the currently willing Provider identities. An observation area and deadline can reveal intended activity even without the full application input. ABE can protect one descriptor for a service-authorized receiver set without listing those identities. [19] Confidential from parties lacking the required decryption capability. Authorized but unselected Providers can read it; names and traffic patterns remain visible.

Slide 12: A Fair Comparison Includes Encryption on Both Sides

Design Execution authorization Discovery confidentiality DNMP-inspired roles [12] Role/command signing rules, credential and request-state checks Add encryption and group/key distribution Our per-service role certificates Service/message rules plus caller or Provider credential Add equivalent encryption and lifecycle handling NDNSF ABE-backed Identity signature, encrypted challenges, and authorization/request state Service-level ABE protection of descriptor Content Trust Schema [16] alone does not encrypt Content. Trust Schema plus encryption is a valid alternative, not an impossible design.

Slide 13: Why Reuse ABE Capability for Authorization?

Enhanced NAC-ABE [19] supports key-policy ABE (KP-ABE): ciphertext attributes meet a policy in an individually issued decryption key (DKEY). Service permissions can be aggregated while identity-signing credentials remain unchanged. Challenges reuse those service permissions instead of introducing additional service-role signing credentials. This saves a second entitlement representation in this design—not all key work. User DKEYs, challenge processing, policy growth, status and rekeying still cost resources.

Slide 14: When Another Authorization Design May Be Better

Known Providers and simple rules: signed invocation plus recipient encryption may be sufficient. A small, stable receiver group: ordinary group-key management may be attractive. Rich command-argument restrictions: DNMP-style rules make those constraints explicit. [12] Many changing entitlements: measure both policy/key updates and credential/rule updates before claiming a management advantage. ABE is a justified option for NDNSF's requirements, not universally superior authorization.

Slide 15: Selected Exchanges and Permission Withdrawal

Symmetric request-scoped keys protect input and result Content. Selection supplies necessary keys only to the selected Provider through recipient encryption. The requesting User and selected Provider share the result key; other Users are excluded under key-secrecy assumptions. Controller-signed status affects authorization after retrieval and validation. ABE rotation limits subsequent encrypted exchanges. Current rotation is Controller-wide. Disconnected revocation is not immediate; old plaintext and keys cannot be recalled. Runtime tests do not establish lower cost than certificate revocation. [28]

Slide 16: Threat Model: What Is Trusted?

Trusted: Controller authorization and the User's planning decision. Untrusted delivery: an object may arrive through a producer, forwarder, cache, or repository and still requires validation. Adversarial cases to test: wrong identities, permissions, challenges, request state, roles, dependencies, or execution identities. Not established by signatures/ABE: honest resource reports, correct inference, truthful images, or resistance to authorized key sharing. No Byzantine-coordinator consensus claim; no traffic-analysis or instantaneous-revocation guarantee.

Slide 17: RQ2: Generate the Plan from Eligible Positive ACKs

1. Application supplies roles, dependency template, constraints and deadline; the template may be handwritten.
2. Close the ACK collection window and validate eligible positive ACKs.
3. A replaceable policy assigns Providers to roles and produces a signed plan.
4. Each Provider derives its permitted input

edges and rechecks local resource admission before starting. The snapshot freezes observed candidates for this decision, not every Provider in the network or their future availability.

Slide 18: A Plan Is More Than a Request ID

Existing request ID Proposed task-specific plan rules Binds messages to a request Assigns exact Providers to collection and processing roles Does not alone specify pre-decessors Declares which producer role may supply each named dependency Does not alone describe key scope Associates permitted objects, execution identity, and authorization/key scope Example: within the same request, a service-authorized UAV that was not assigned “view 1” must not satisfy the view-1 dependency.

Slide 19: Plan-to-Data Enforcement: Check Before Advancing

1. Validate the Data signature using the appropriate trust rules. 2. Match the current request, plan and execution identity. 3. Verify that the signer is the assigned predecessor Provider. 4. Match the declared dependency edge, name, data schema and key scope. 5. Require complete validated input before the dependent computation starts. Apply the same rules to direct, cached and repository-fetched Data. This is a proposed acceptance contract; full end-to-end qualification remains required.

Slide 20: Failure Handling Needs Explicit State

Event Required behavior Late ACK Do not silently change the committed plan Duplicate Selection Follow recorded one-time state; do not implicitly repeat a side effect Incomplete dependency Wait within policy/deadline, recover, or fail; do not execute on partial data Provider replacement Authorize a new assignment; define fencing and whether existing results may be reused Cancellation Prevent late messages from reauthorizing cancelled work Network re-retrieval is not task re-execution. Compensation is not an atomic transaction or exactly-once physical action.

Slide 21: Proposed UAV Flow: Capture, Analyze and Return

1. Sensing UAVs publish encrypted ViewImage Data in their own authorized namespaces. 2. The compute Provider retrieves images over assigned edges and publishes Annotation Data. 3. The result references the original images and securely provides permitted image keys to the requester. 4. The requester retrieves the original images and displays the annotations. No need to republish the same images as if the compute Provider produced them. GPS alone does not establish camera visibility or accurate triangulation.

Slide 22: What the UAV Case Must Demonstrate

A complete authorized collection–processing–return task before the deadline. Correct handling of unavailable views, delayed images and a failed processing Provider. Wrong-role/edge rejection without rejecting an authorized cached image. MiniNDN [22] and PX4 SITL [23] results clearly distinguished from physical-flight evidence. Higher recognition/localization accuracy is a separate claim requiring ground truth and a single-view control. It is not assumed here.

Slide 23: Proposed DI Flow: Compute Follows Named Dependencies

Provider A layer group 1 Provider B layer group 2 Provider C output stage activation activation Pipeline: sequential layers/groups; tensor partitioning additionally needs supported collectives. ONNX [25] describes the graph; export alone does not implement distributed execution. Repo [24] stores artifacts; forwarding caches do not guarantee persistence. The framework contribution is checked runtime assignment and data exchange, not model partitioning itself.

Slide 24: DI State and Evidence Boundaries

Model weights, reusable artifacts, request activations and KV state have different lifetimes. Autoregressive decoding requires retained K/V and next-step inputs; resident weights do not remove that state dependency. Native/local tests are not cross-process or multi-device qualification. Freeze one supported model/partition first; tensor/hybrid and LLM streaming need their own evidence if claimed. Earlier small-model measurements do not qualify all current V3/native features. [28]

Slide 25: RQ3: Define the Baselines Before Comparing

NSC [10]: named service calls; describe the tested executor, timeout, queue and recovery behavior. gRPC [17]: describe resolver, sequential/parallel policy, deadlines, retry and cancellation. Match Provider count, capacity, task, topology/traces, publication times and measurement windows. Report fixed-list and dynamically discovered configurations as different configurations. An incomplete baseline cannot establish a limitation of the full framework. Historical unmatched data remain diagnostics.

Slide 26: Mobility: Overall Completion Depends on Coverage

Ten seeds (62–71), four Providers/system, 2 m/s; mean seed completion rate. [28] AP radius NDNFSF gRPC NSC 50 m 54.57% 53.80% 55.47% 100 m 97.97% 97.97% 98.33% 150 m 100.00% 100.00% 100.00% These overall controls do not establish NDNFSF superiority. The next slide examines the separately registered switching subset, not all requests.

Slide 27: Mobility: Lower Tail Latency in a Defined Subset

Ten seeds; four Providers/system; 100 m, 2 m/s; 1,312 matched requests requiring a Provider switch. [28] Comparison Seed-p95 difference 95% CI NDNFSF minus gRPC-SEQ-4 -597 ms [-992, -200] ms NDNFSF minus NSC-4 -2618 ms [-2916, -2315] ms This is a conditional latency result, not general completion-rate superiority. Preserve the registered subset definition, overall coverage controls, and failed requests.

Slide 28: Selection Avoids Redundant Starts in This Comparison

Six-seed, capacity-matched comparison; 1,800 requests per system. [28] Configuration Completed requests Execution starts NDNFSF 1,798 / 1,800 1,800 gRPC-PAR-4 1,800 / 1,800 7,180 NSC-4 also completed 1,800/1,800, with 1.002 starts/request (rounded). No work-efficiency advantage over NSC is claimed. Fewer starts than gRPC-PAR-4 is not proof of fewer bytes, less CPU time or lower energy. Similar observed completion is not proven equivalence.

Slide 29: A Useful Negative Result: Splitting Can Be Slower

Earlier small Qwen-style ONNX model; ten successful runs per layout. [28] Layout Mean latency p95 Shared-backbone distributed layout 259.4 ms 283.9 ms Single Provider 186.8 ms 206.3 ms Do not split merely because several Providers exist. Communication/setup overhead is a candidate explanation, not a causal result established by this measurement. This campaign does not validate the newer canonical-layer, tensor/hybrid or multi-accelerator design.

Slide 30: Remaining Experiments Follow the Claims

RQ Required evidence Outcome and cost RQ1 Full authorization alternatives; unauthorized call, challenge/replay and withdrawal cases No unauthorized handler start; discovery delay, key/policy size, status delay and rekey traffic RQ2 One UAV and one DI graph; wrong role/edge; direct/cache parity; cancellation/replacement Correct task completion; validation, transfer and recovery cost RQ3 Matched workload and failure schedules; feasible single-device control; cold/warm state Completion plus latency, redundant work, bytes, uncertainty, and cases with no advantage Planned tests are not completed findings. Record all scheduled runs and failure boundaries.

Slide 31: A Bounded Path to Spring 2027

Internal target Completion gate September 2026 Agree scope/RQs; freeze one UAV graph and one DI partition; resolve correctness blockers October–November 2026 Complete registered experiments; freeze evidence and write bounded RQ answers December 2026 Complete integrated advisor-review draft January 2027 Committee review and readiness decision; reduce scope or revise timing if core evidence is missing February 2027 Target defense, conditional on evidence and committee approval CS guidance: advisor-approved complete draft to committee two weeks before defense. Spring 2027 final ProQuest deadline: March 26, 2027; recheck before filing. [29,30]

Slide 32: What a Final Defense Must Be Able to Claim

RQ1 answer: when ABE-backed discovery/authorization is preferable, including management and revocation costs. RQ2 answer: exactly which task-assignment mistakes are rejected and which recovery behavior is supported. RQ3 answer: measured benefits, negative results and deployment limits. Distinguish the candidate’s research from shared implementations and reused NDN/AI libraries. This proposal defines those obligations. It does not label the remaining collaboration evidence as finished.

Slide 33: Discussion: Is This Scope Sufficient and Feasible?

1. Are the three research questions specific enough to support defensible answers? 2. Are the authorization alternatives complete and fairly configured? 3. Are one UAV graph and one supported DI partition sufficient to test the abstraction? 4. Which optional features should be excluded if they delay the core evidence? NDN SF: runtime service discovery and authorization, connected to explicit named-data execution rules.

Slide 34: Backup: Discovery Configuration Is Not Framework Capability

Provider-transition configuration Completed NDN SF, no pre-registered Provider addresses 357 / 357 Static-list NSC/gRPC configurations 0 / 357 Pre-registered baseline variants 357 / 357 This isolates the configured discovery boundary. Resolver/discovery integration can update candidates; it is not proof that gRPC or NSC cannot do so. [27,28]

Slide 35: Backup: Historical Loss Evidence Is Not a New Matched Test

Historical per-link loss NDN SF completed requests 1% 599 / 600 3% 598 / 600 10% (best recorded run) 539 / 600 A selected run is not an aggregate estimate. Do not combine these numbers with another batch and label the result “matched.” [28] 1 - (1 - ℓ)⁴ assumes four independent equal-loss packet traversals; it is not service failure probability after retries.

Slide 36: References: NDN and Authorization

[1] L. Zhang et al., “Named Data Networking,” ACM SIGCOMM CCR, 2014. [2] T. Li et al., “A Brief Introduction to NDN Dataset Synchronization (NDN Sync),” MILCOM, 2018. [3] P. Moll et al., “A Brief Introduction to State Vector Sync,” NDN-0073, rev. 2, 2021. [4] Z. Zhang et al., “NAC: Automating Access Control via Named Data,” MILCOM, 2018. [12] K. Nichols, “Lessons Learned Building a Secure Network Measurement Framework using Basic NDN,” ACM ICN, 2019. [16] Y. Yu et al., “Schematizing Trust in Named Data Networking,” ACM ICN, 2015. [19] S. Dulal et al., “Enhancing NAC-ABE to Support Access Control for mHealth Applications and Beyond,” arXiv:2311.07299, 2023. [26] NDN Project, “Data Packet,” NDN Packet Format Specification. docs.named-data.net/NDN-packet-spec/current/data.html (accessed Sept. 2026).

Slide 37: References: Invocation and Composition

[6] C. Tschudin and M. Sifalakis, “Named Functions and Cached Computations,” IEEE CCNC, 2014. [7] M. Krol and I. Psaras, “NFaaS: Named Function as a Service,” ACM ICN, 2017. [9] M. Krol et al., “RICE: Remote Method Invocation in ICN,” ACM ICN, 2018. [10] D. Meirovitch and L. Zhang, “NSC—Named Service Calls, or a Remote Procedure Call for NDN,” NDN-0074, rev. 1, 2021. [15] CaSCON, SMARTCOMP 2025. Primary publication record: RIT, “Barrios presents at IEEE SmartComp.” Full feature-level comparison remains a research obligation. [17] gRPC Authors, “Core Concepts, Architecture and Lifecycle,” gRPC documentation. [27] gRPC Authors, “Custom Name Resolution,” grpc.io/docs/guides/custom-name-resolution/ (accessed Sept. 2026).

Slide 38: References: Tools, Evidence and Degree Planning

[22] Mini-NDN, project documentation: minindn.ndn.today. [23] PX4, “Software-in-the-Loop Simulation,” PX4 User Guide. [24] NDN Project, “Repo Protocol Specification,” repo-ng documentation. [25] ONNX, “Open Neural Network Exchange Intermediate Representation Specification.” [28] NDN SF, research-revision-audit.md, companion claim/evidence ledger, September 9, 2026. Contains campaign paths and unresolved acceptance gates. [29] University of Memphis CS, “Information for Computer Science Ph.D. Students,” forms_grad_phd_programinfo.pdf (accessed September 9, 2026). [30] University of Memphis Graduate School, “Graduation Information,” Spring 2027 calendar (accessed September 9, 2026).